

torch.linalg.eigvals#

<https://pytorch.org/docs/stable/generated/torch.linalg.eigvals.html>

Description:

Computes the eigenvalues of a square matrix. Letting $K \in \mathbb{R}^{n \times n}$ or $C \in \mathbb{C}^{n \times n}$, the eigenvalues of a square matrix $A \in K^{n \times n}$ are defined as the roots (counted with multiplicity) of the polynomial p of degree n given by $p(\lambda) = \det(A - \lambda I_n)$ where I_n is the n -dimensional identity matrix. Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if A is a batch of matrices then the output has the same batch dimensions. The returned eigenvalues are not guaranteed to be in any specific order.

Function Signature

```
torch.linalg.eigvals(A, *, out=None) → Tensor#
```

Parameters

Keyword Arguments

Returns

Parameters

Keyword

out (Tensor, optional) – output tensor. Ignored if None. Default: None.

Returns:

A complex-valued tensor containing the eigenvalues even when A is real.

Code Examples

```
>>> A = torch.randn(2, 2, dtype=torch.complex128)
>>> L = torch.linalg.eigvals(A)
>>> L
tensor([ 1.1226+0.5738j, -0.7537-0.1286j],
        dtype=torch.complex128)

>>> torch.dist(L, torch.linalg.eig(A).eigenvalues)
tensor(2.4576e-07)
```

Notes & Warnings

NOTE

Note The eigenvalues of a real matrix may be complex, as the roots of a real polynomial may be complex. The eigenvalues of a matrix are always well-defined, even when the matrix is not diagonalizable.

NOTE

Note When inputs are on a CUDA device, this function synchronizes that device with the CPU.

NOTE

See also `torch.linalg.eig()` computes the full eigenvalue decomposition.

Detailed Documentation

Documentation

Computes the eigenvalues of a square matrix.

Letting $K \in \mathbb{R} \text{ or } \mathbb{C}$, the eigenvalues of a square matrix $A \in K^{n \times n}$ are defined as the roots (counted with multiplicity) of the polynomial p of degree n given by

where I_n is the n -dimensional identity matrix.

Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of

matrices, and if A is a batch of matrices then the output has the same batch dimensions.

The returned eigenvalues are not guaranteed to be in any specific order.

The eigenvalues of a real matrix may be complex, as the roots of a real polynomial may be complex.

The eigenvalues of a matrix are always well-defined, even when the matrix is not diagonalizable.

When inputs are on a CUDA device, this function synchronizes that device with the CPU.

`torch.linalg.eig()` computes the full eigenvalue decomposition.

A (Tensor) – tensor of shape $(*, n, n)$ where $*$ is zero or more batch dimensions.

out (Tensor, optional) – output tensor. Ignored if `None`. Default: `None`.

A complex-valued tensor containing the eigenvalues even when A is real.